Advanced Graphics in PDF

This PDF showcases complex graphics generated with Python's Pillow library, including gradients, fractals, charts, and geometric patterns.

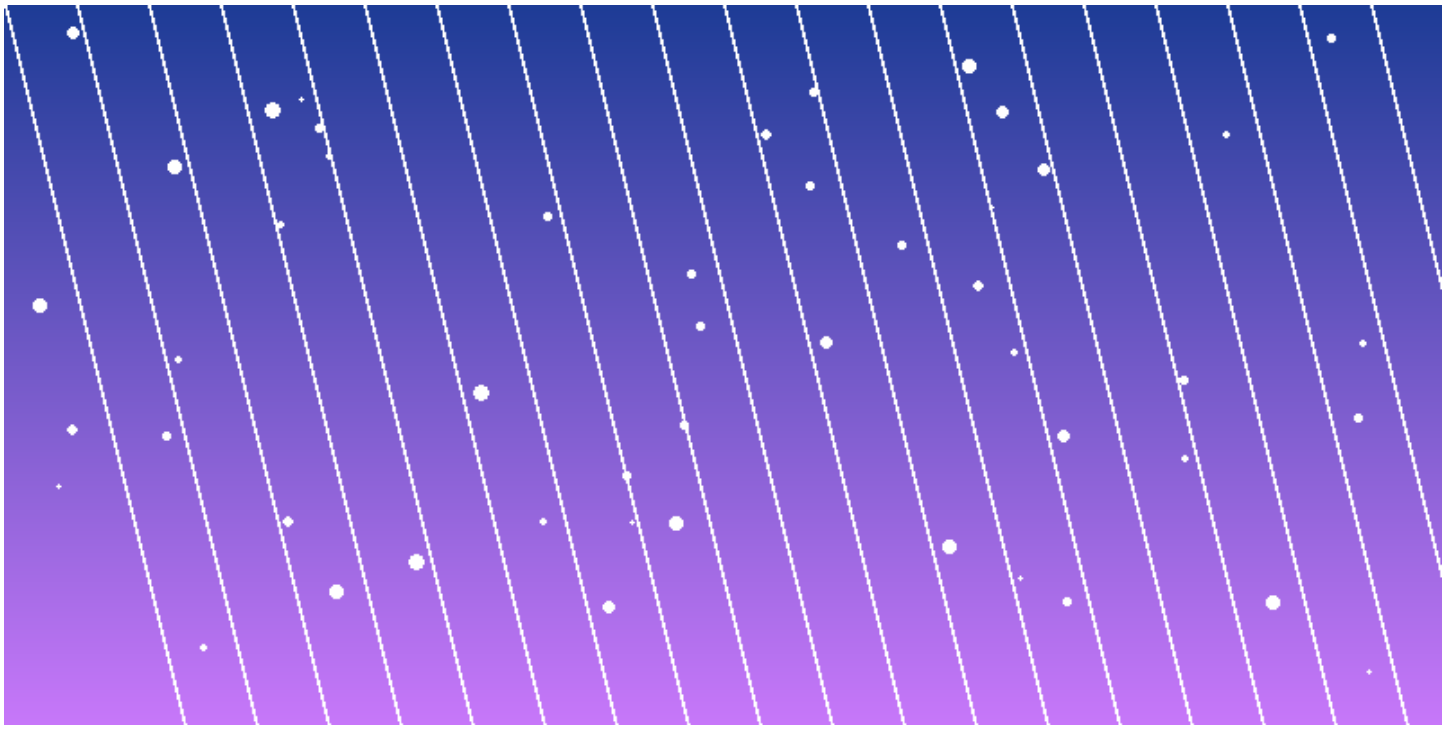


Figure 1: Gradient background with overlay patterns

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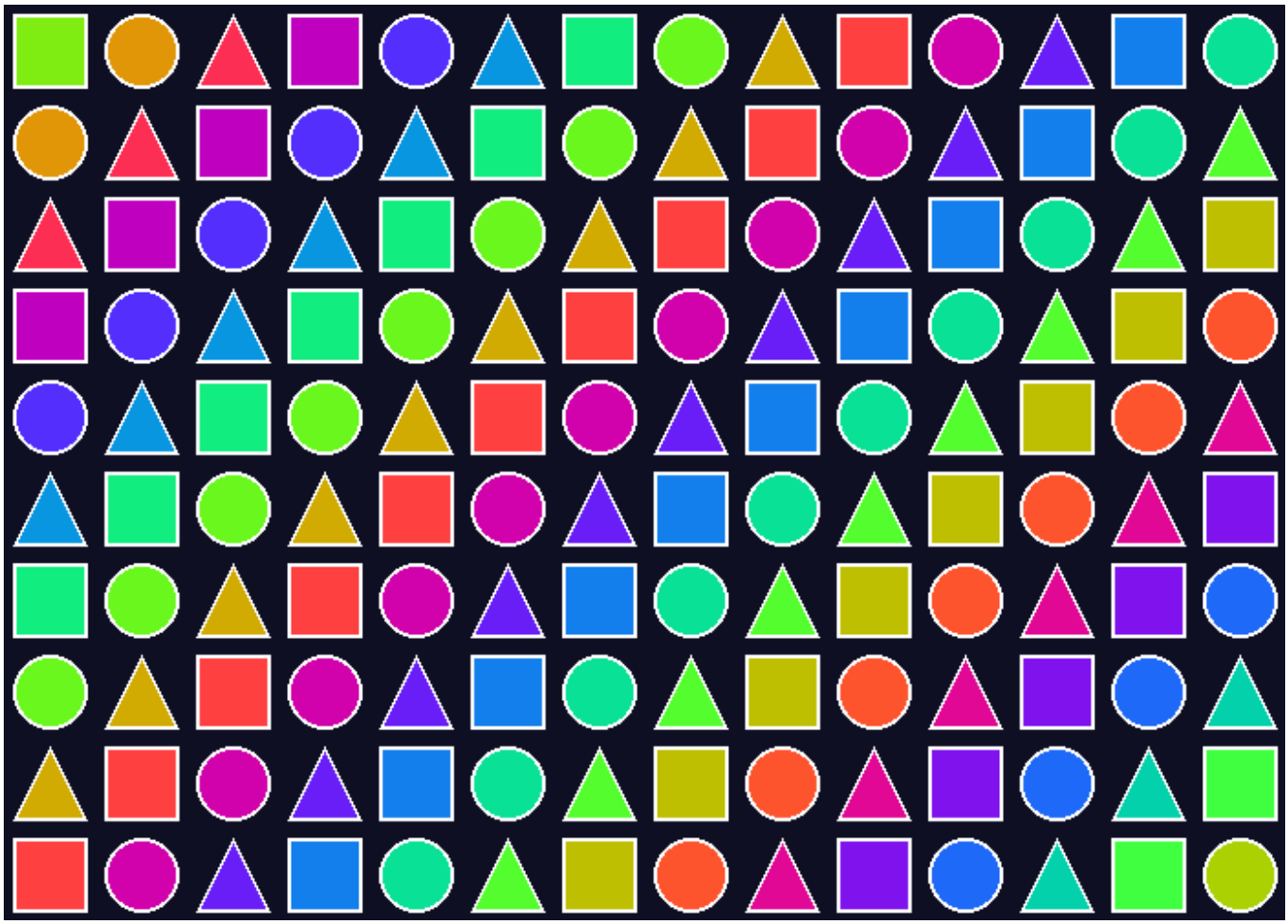


Figure 2: Geometric tessellation with dynamic colors

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Fractal Mathematics

The Mandelbrot set is a famous fractal generated by iterating the formula z = z² + c. Each pixel's color represents how quickly the iteration diverges.

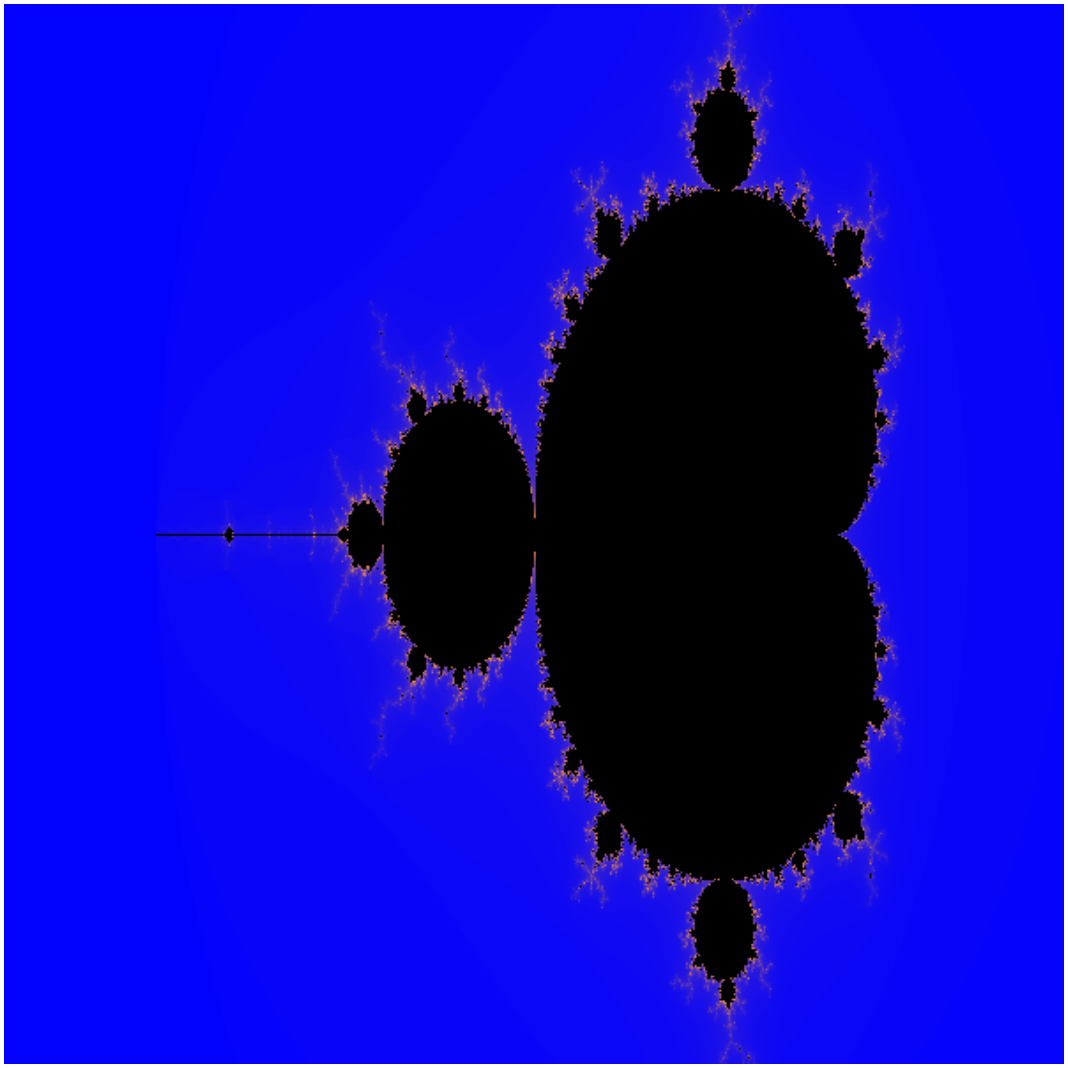


Figure 3: Mandelbrot set visualization (100 iterations)

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Data Analytics Dashboard

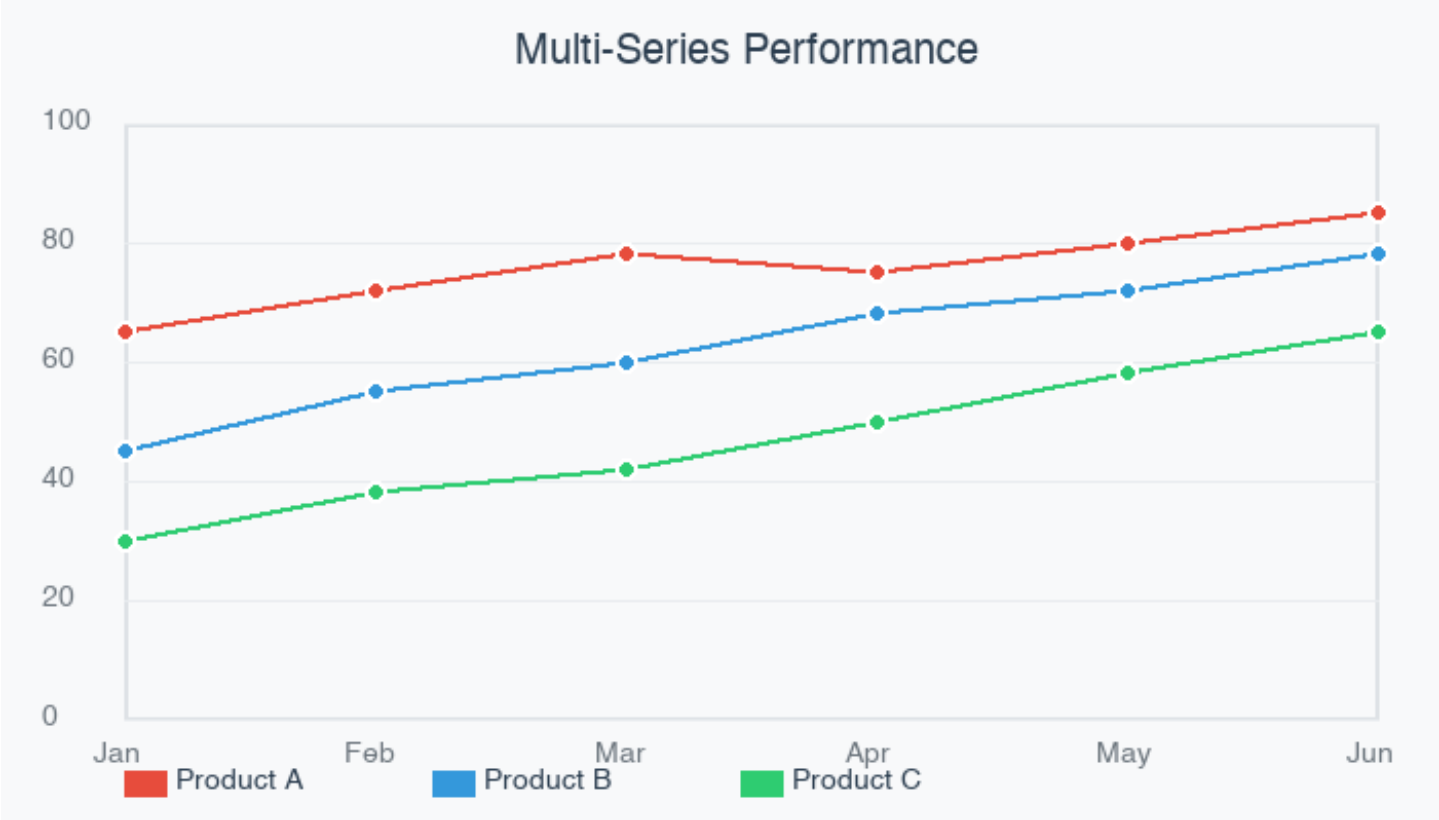


Figure 4: Multi-series line chart with interactive-style design

The chart above demonstrates multiple data series with distinct colors, data point markers, grid lines, and a legend - all generated programmatically.

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Visual Patterns

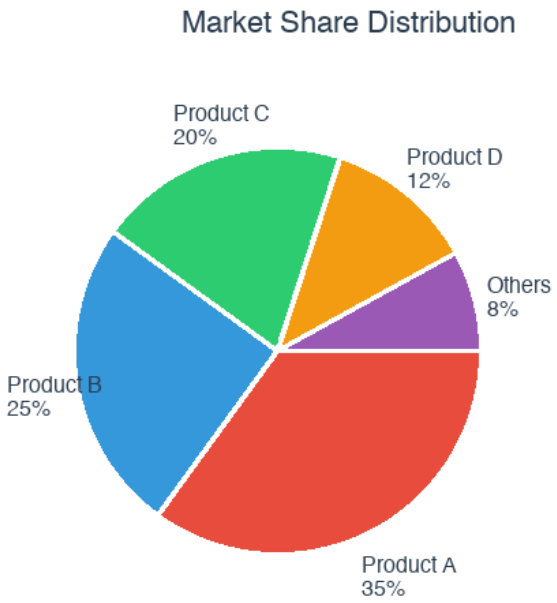


Figure 5: Pie chart with labels



Figure 6: Spiral pattern with RGB color cycling

Left: Market share distribution shown as a pie chart with percentage labels. Right: A mathematical spiral pattern where colors cycle through the RGB spectrum based on angle.